Chapter 34

Vertebrates

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: Half a Billion Years of Backbones

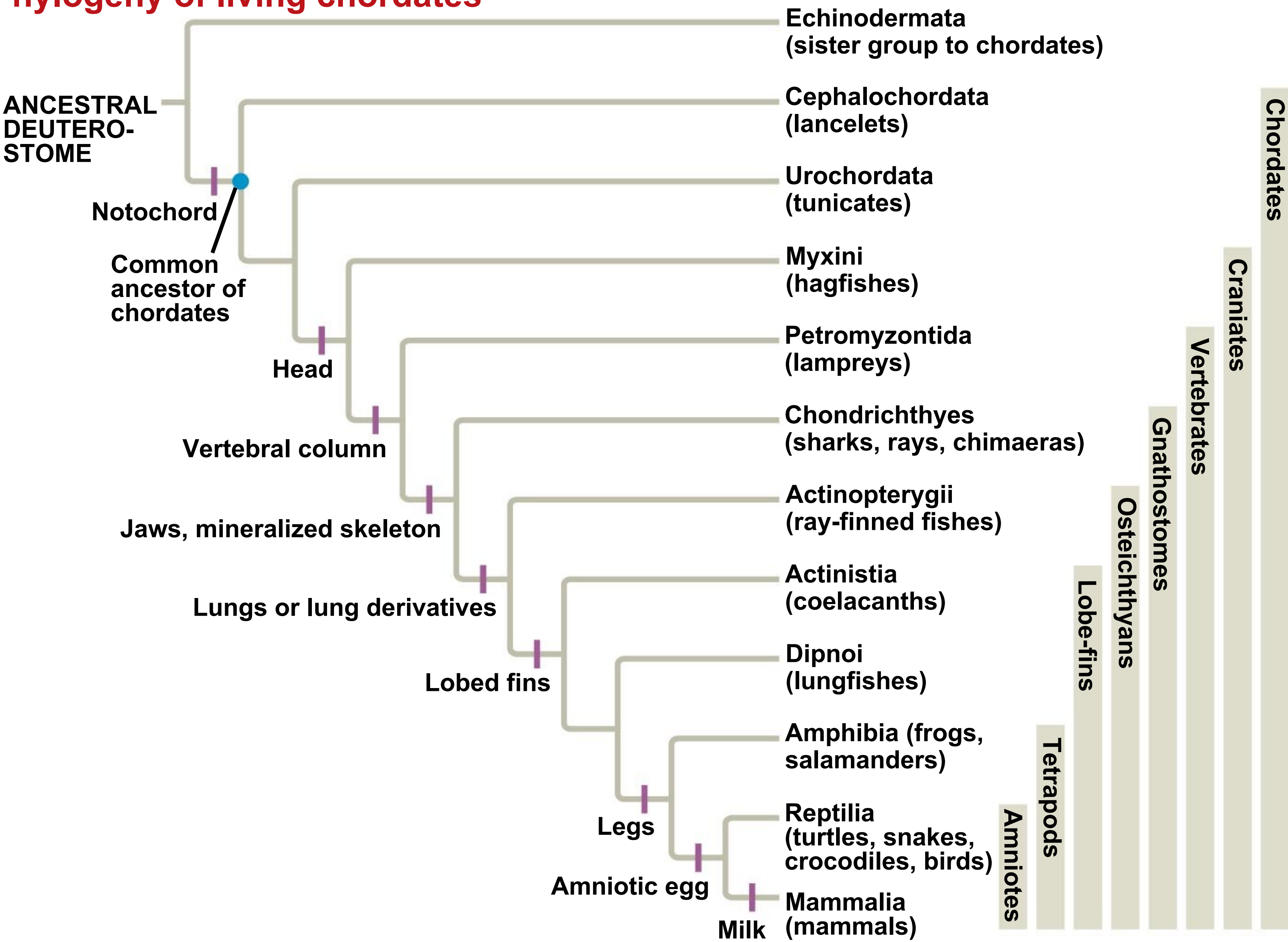
- Early in the Cambrian period, about 530 million years ago, an astonishing variety of animals inhabited Earth's oceans.
- One type of animal gave rise to vertebrates, one of the most successful groups of animals.
- The animals called **vertebrates** get their name from vertebrae, the series of bones that make up the backbone.
- There are about 52,000 species of vertebrates, including the largest organisms ever to live on the Earth.

Are humans among the descendants of this ancient organism?

Concept 34.1: Chordates have a notochord and a dorsal, hollow nerve cord

- Vertebrates are a subphylum within the phylum Chordata.
- **Chordates** are bilaterian animals that belong to the clade of animals known as Deuterostomia.
- Two groups of invertebrate deuterostomes, the urochordates and cephalochordates, are more closely related to vertebrates than to other invertebrates.

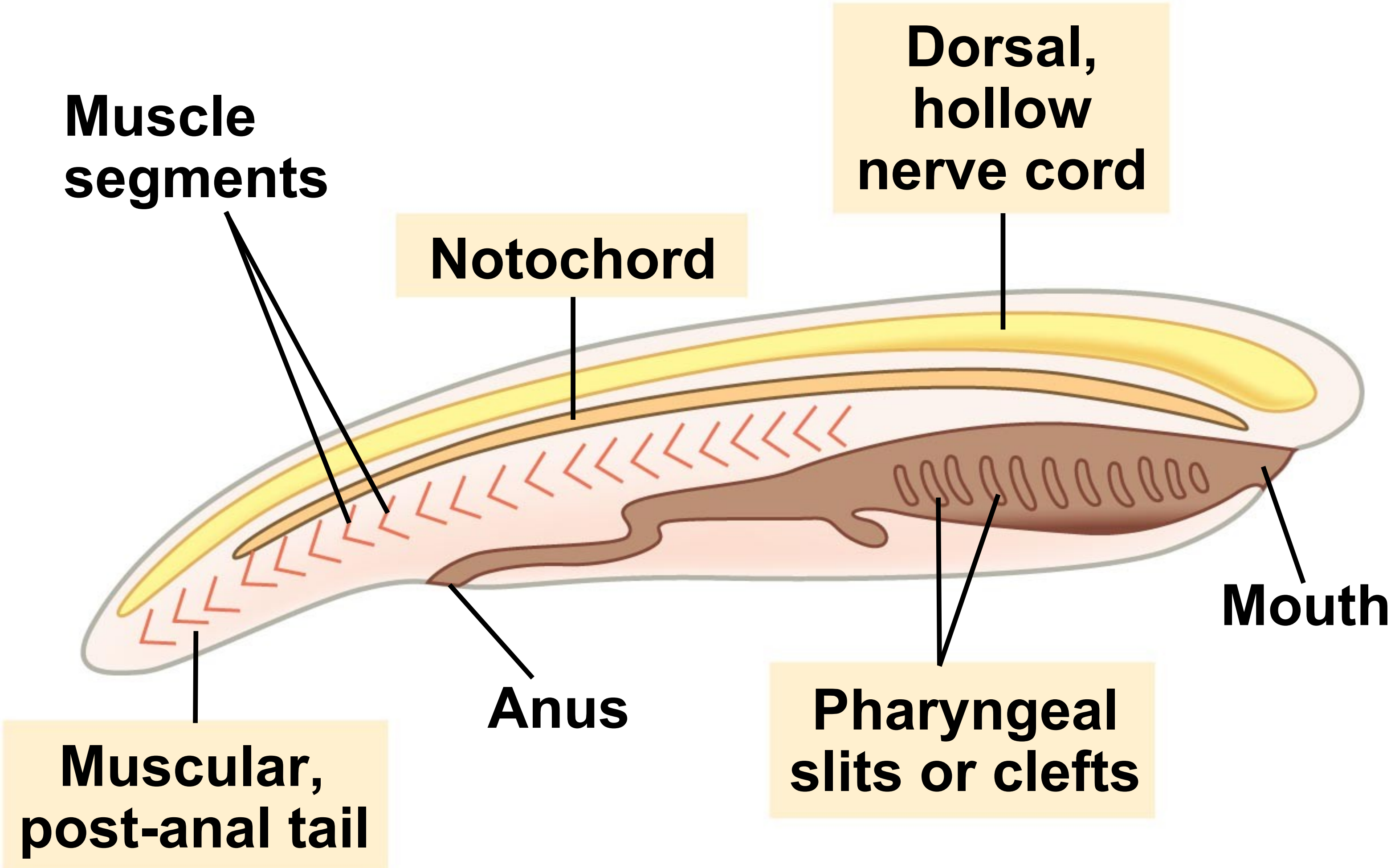
Phylogeny of living chordates



Derived Characters of Chordates

- All chordates share a set of derived characters.
- Some species have some of these traits only during embryonic development.
- ***Four key characters of chordates:***
  - ***Notochord***
  - ***Dorsal, hollow nerve cord***
  - ***Pharyngeal slits or clefts***
  - ***Muscular, post-anal tail***

Chordate characteristics



Notochord

- The **notochord** is a longitudinal, flexible rod between the digestive tube and nerve cord.
- It provides skeletal support throughout most of the length of a chordate.
- In most vertebrates, a more complex, jointed skeleton develops, and the adult retains only remnants of the embryonic notochord.

Dorsal, Hollow Nerve Cord

- The nerve cord of a chordate embryo develops from a plate of ectoderm that rolls into a tube dorsal to the notochord.
- The nerve cord develops into the central nervous system: the brain and the spinal cord.

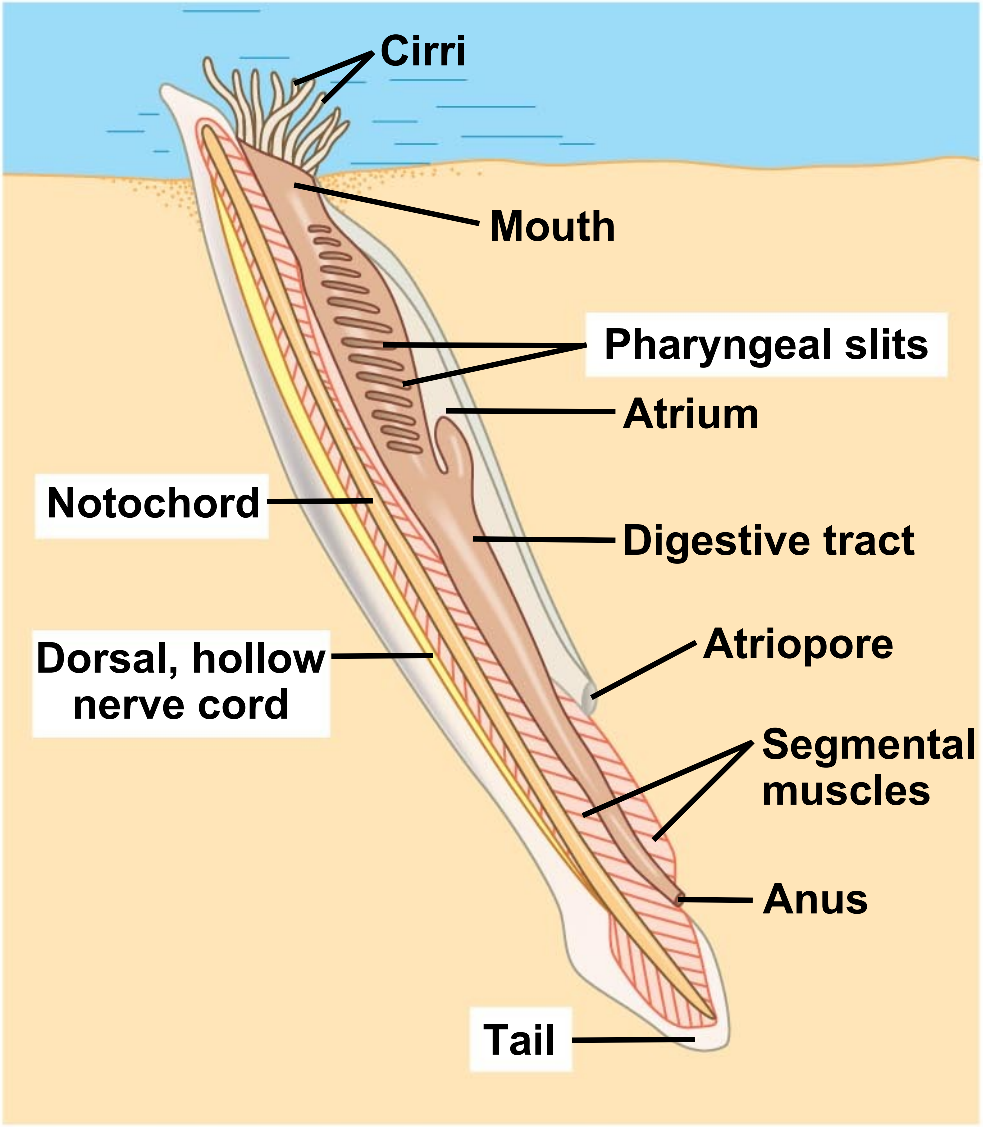
Pharyngeal Slits or Clefts

- In most chordates, grooves in the pharynx called **pharyngeal clefts** develop into slits that open to the outside of the body.
- Functions of **pharyngeal slits**:
  - Suspension-feeding structures in many invertebrate chordates
  - Gas exchange in vertebrates (except vertebrates with limbs, the tetrapods)
  - Develop into parts of the ear, head, and neck in tetrapods.

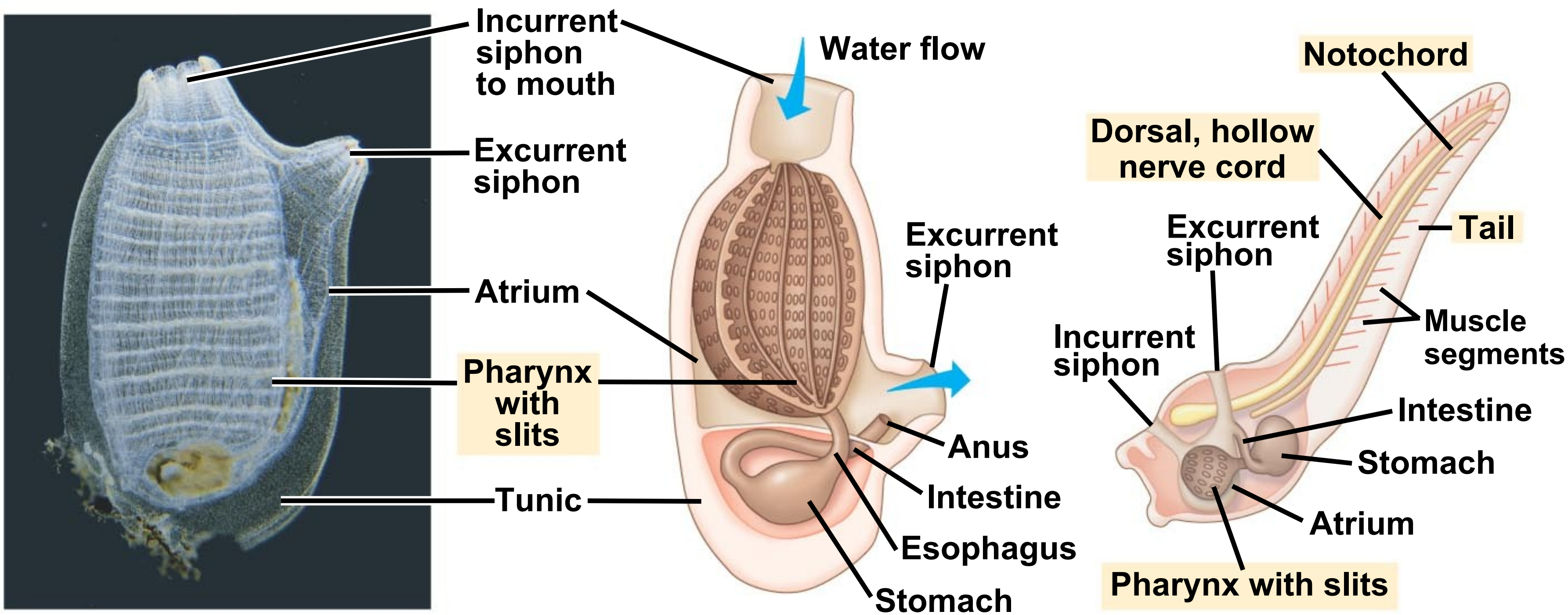
Muscular, Post-Anal Tail

- Chordates have a tail posterior to the anus.
- In many species, the tail is greatly reduced during embryonic development.
- The tail contains skeletal elements and muscles.
- It provides propelling force in many aquatic species.

Lancelets are named for their bladelike shape. They are marine suspension feeders. Adults retain characteristics of chordate body plan.



Tunicates (Urochordata) are more closely related to other chordates than are lancelets. They are marine suspension feeders commonly called sea squirts. As an adult, a tunicate draws in water through an incurrent siphon, filtering food particles. Juveniles, not adults, have a notochord.



An adult tunicate

A tunicate larva

Fossil of an early Chordate

Hagfishes have a cartilaginous skull and axial rod of cartilage derived from the notochord, but lack jaws and vertebrae

Derived Characters of Vertebrates

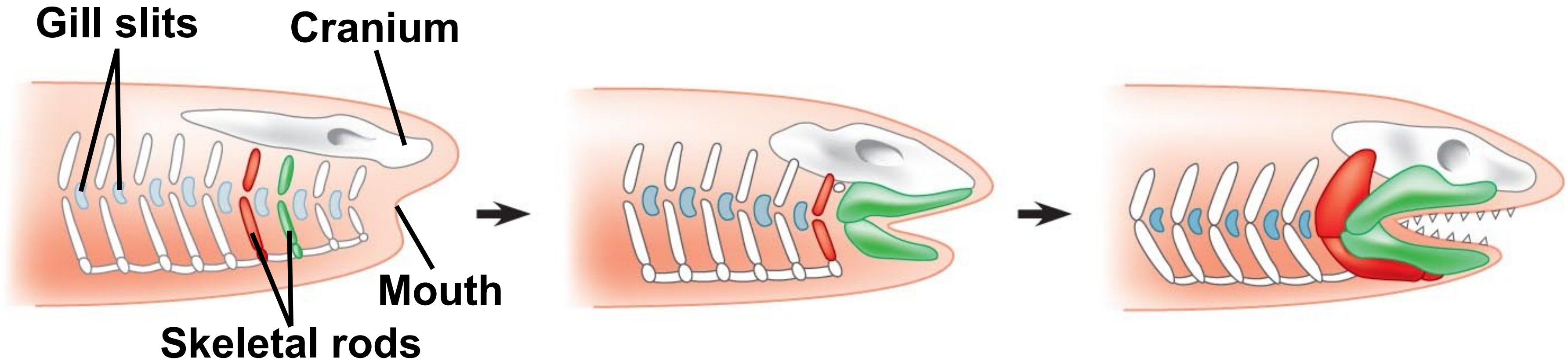
- During the Cambrian period, a lineage of craniates evolved into vertebrates. Vertebrates became more efficient at capturing food and avoiding being eaten.
- Vertebrates have the following derived characters:
  - Vertebrae enclosing a spinal cord
  - An elaborate skull
  - Fin rays, in the aquatic forms.

Lampreys represent the oldest living lineage of vertebrates. They are jawless vertebrates inhabiting various marine and freshwater habitats.

Origins of Bone and Teeth

- Mineralization appears to have originated with vertebrate mouthparts.
- The vertebrate endoskeleton became fully mineralized much later.
- Today, *jawed vertebrates*, or ***gnathostomes***, outnumber jawless vertebrates.
- Gnathostomes jaws might have evolved from skeletal supports of the pharyngeal slits.

Hypothesis for the evolution of vertebrate jaws

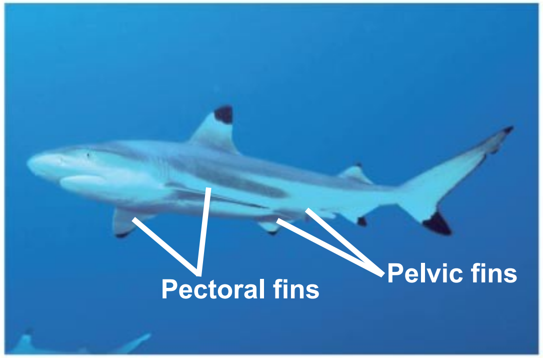


- Other characters common to gnathostomes:
  - An additional duplication of **Hox genes**
  - An enlarged forebrain associated with enhanced smell and vision
  - In aquatic gnathostomes, the **lateral line system**, which is sensitive to vibrations.

Chondrichthyans (Sharks, Rays, and Their Relatives)

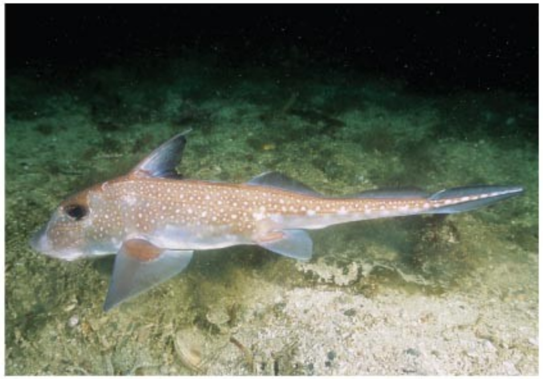
- **Chondrichthyans** (Chondrichthyes) have a skeleton composed primarily of cartilage.
- The cartilaginous skeleton evolved secondarily from an ancestral mineralized skeleton.
- The largest and most diverse group of chondrichthyans includes the sharks, rays, and skates.

Chondrichthyans



(a) Blacktip reef shark (*Carcharhinus melanopterus*)

(b) Southern stingray (*Dasyatis americana*)



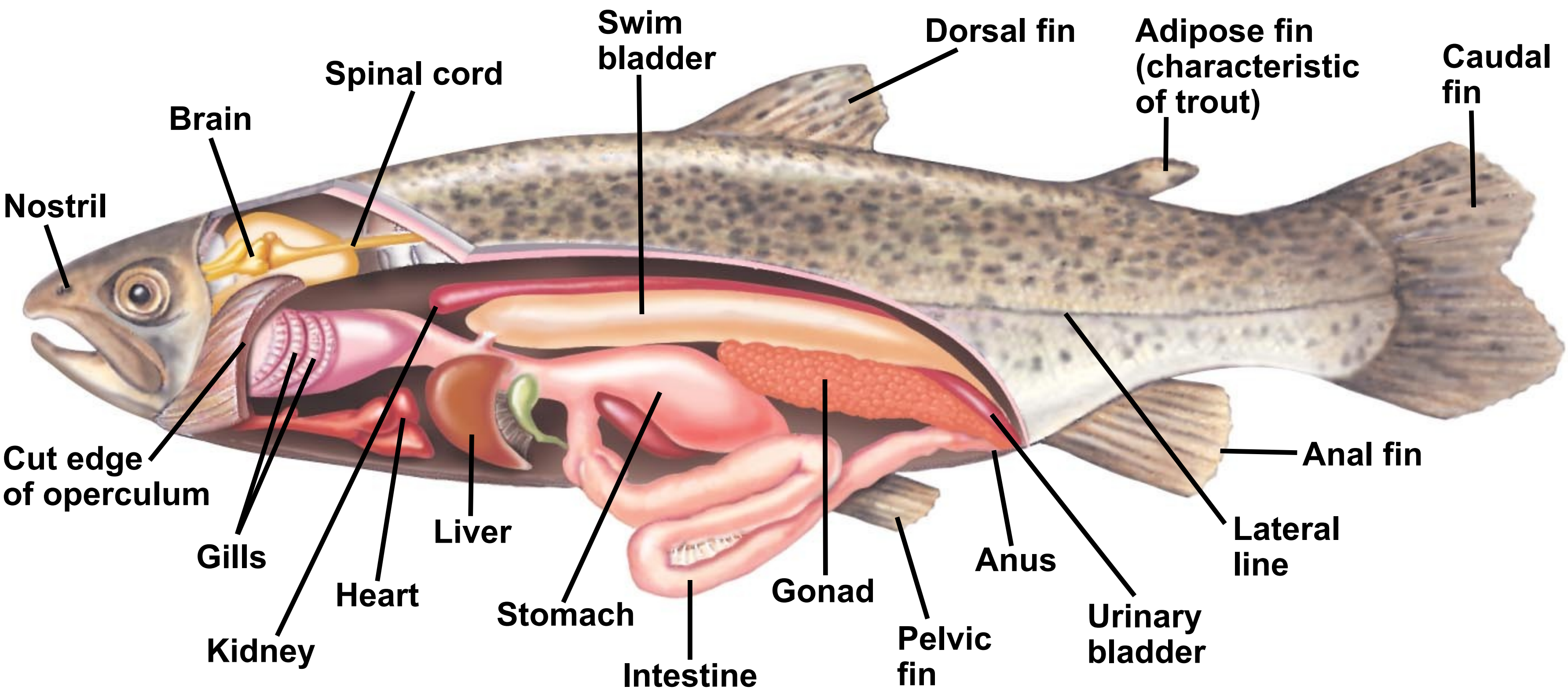
(c) Spotted ratfish (*Hydrolagus colliei*)

- Most sharks
  - Have a streamlined body and are swift swimmers
  - Are carnivores
  - Have a short digestive tract; a ridge called the *spiral valve* increases the digestive surface area
  - Have acute senses.

- Shark eggs are fertilized internally but embryos can develop in different ways:
  - **Oviparous**: eggs hatch outside the mother's body.
  - **Ovoviviparous**: the embryo develops within the uterus and is nourished by the egg yolk.
  - **Viviparous**: the embryo develops within the uterus and is nourished through a yolk sac placenta from the mother's blood.
- The reproductive tract, excretory system, and digestive tract empty into a common **cloaca.**

- The vast majority of vertebrates belong to a clade of gnathostomes called Osteichthyes.
- ***Osteichthyes*** includes the bony fish and tetrapods.
- They have a *bony endoskeleton.*
- Aquatic osteichthyans are the vertebrates we informally call fishes.
- Most fishes breathe by drawing water over gills protected by an **operculum.**
- Fishes control their buoyancy with an air sac known as a **swim bladder.**

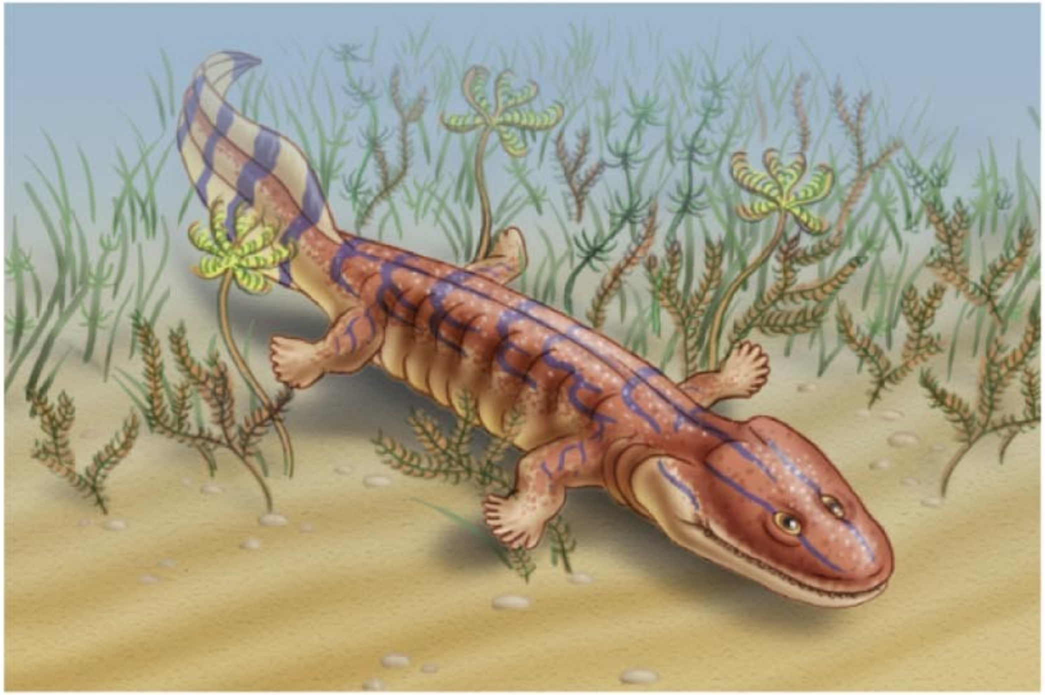
Anatomy of a trout - bony fish - Osteichthyes

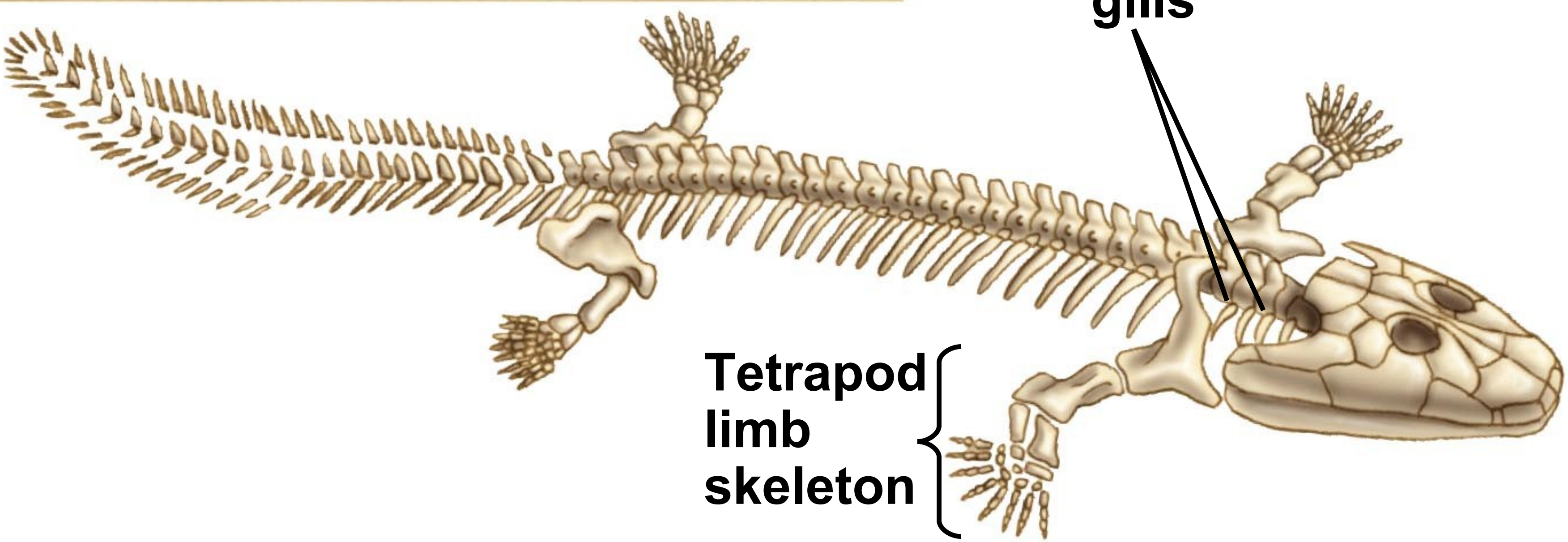


Derived Characters of Tetrapods

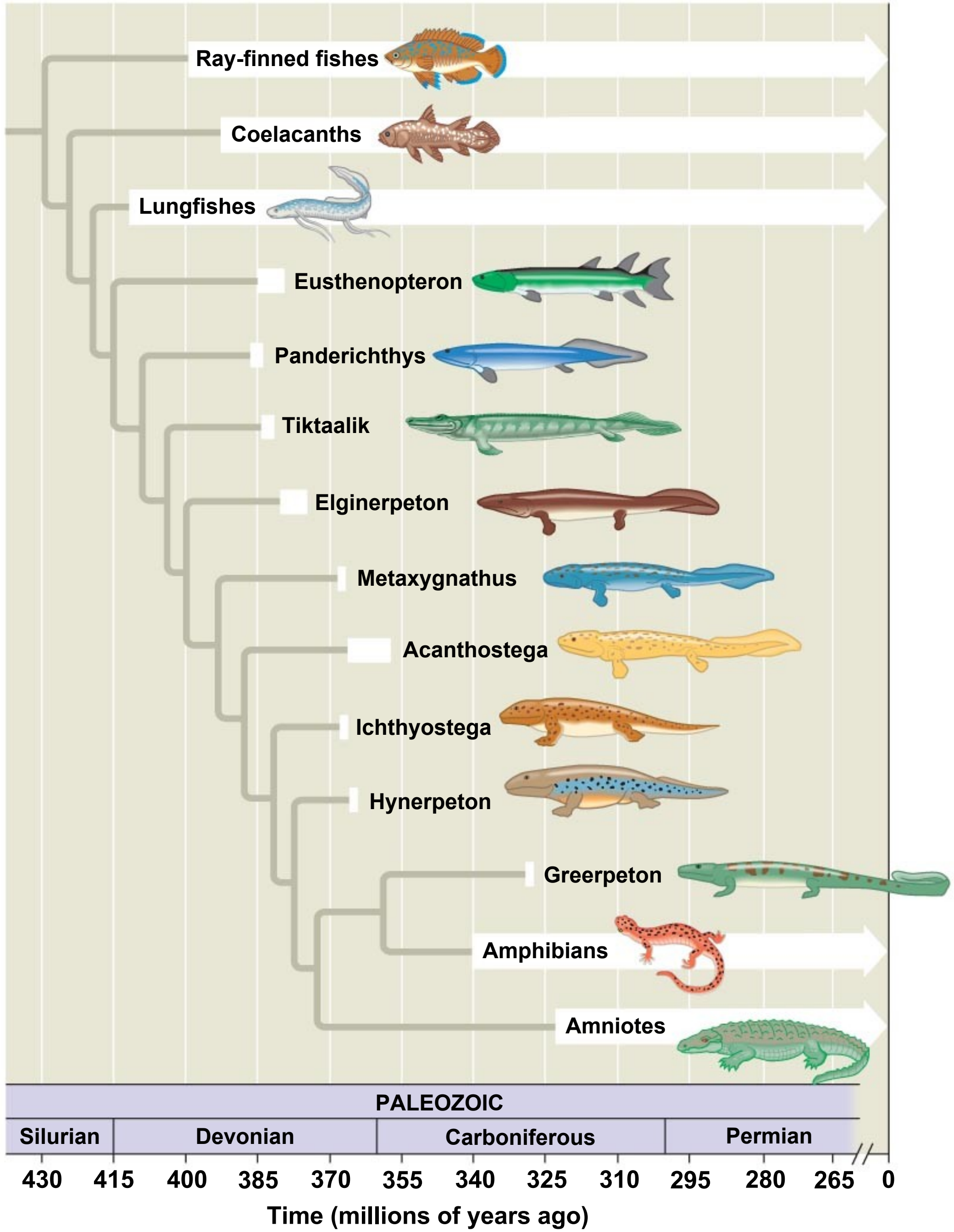
- **Tetrapods** have some specific adaptations:
  - Four limbs, and feet with digits
  - Ears for detecting airborne sounds.
- In one lineage of lobe-fins, the fins became progressively more limb-like while the rest of the body retained adaptations for aquatic life.
- For example, *Acanthostega* lived in Greenland 365 million years ago.

A Devonian era relative of tetrapods





Origin of Tetrapods



Amphibians

- **Amphibians** (class Amphibia) are represented by about 6,150 species of organisms in three orders.
- *Amphibian* means “both ways of life,” referring to the metamorphosis of an aquatic larva into a terrestrial adult.
- Most amphibians have moist skin that complements the lungs in gas exchange.
- Fertilization is external in most species, and the eggs require a moist environment.

Amphibians

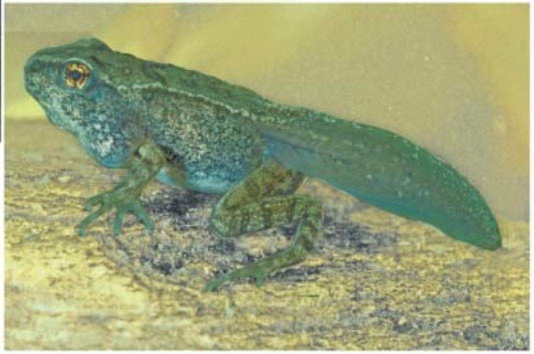
(a) Order Urodela

(b) Order Anura

(c) Order Apoda

The “dual life” of a frog

(a) Tadpole



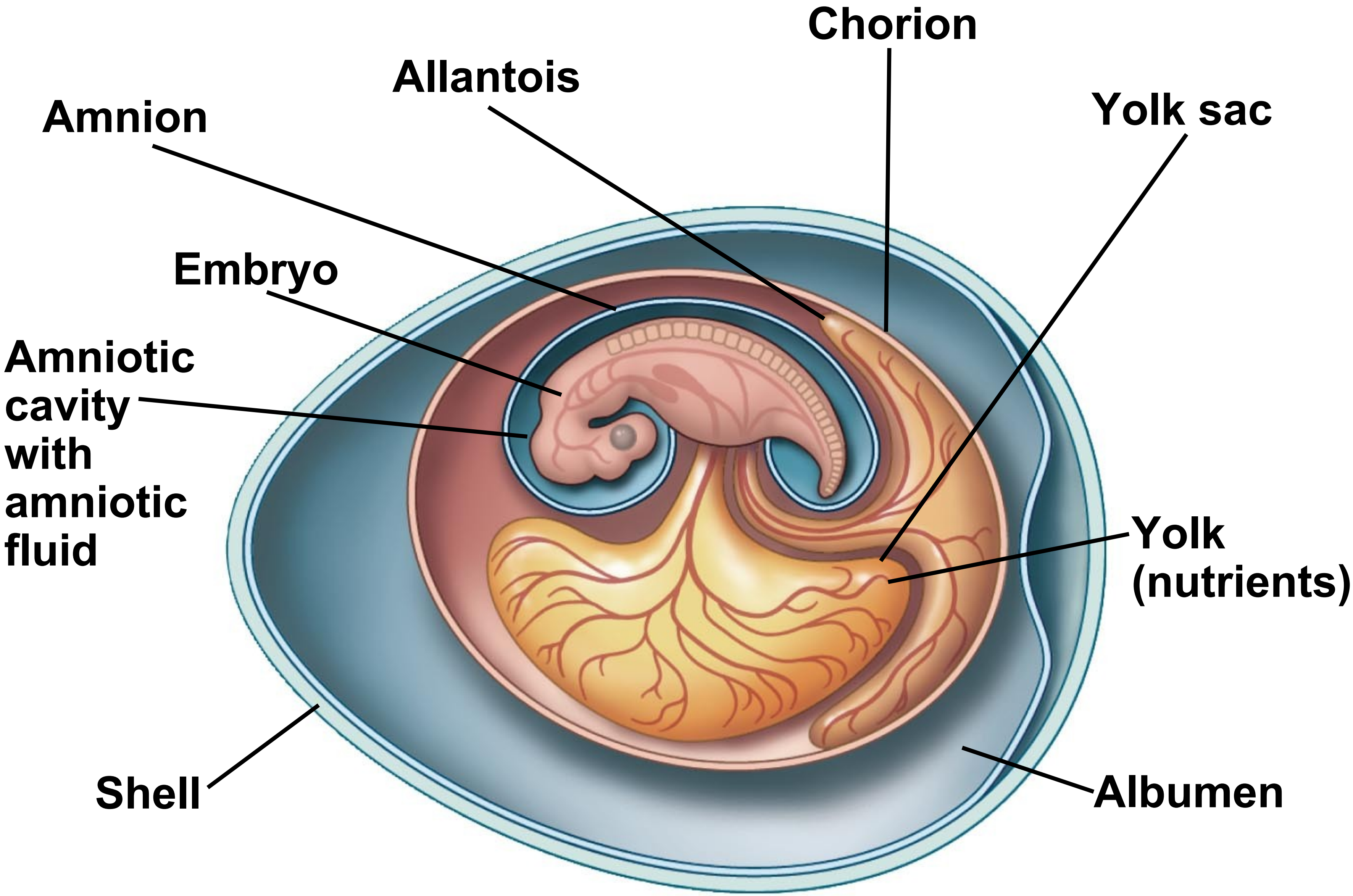
(b) During metamorphosis

(c) Mating adults

Concept 34.6: Amniotes are tetrapods that have a terrestrially adapted egg

- **Amniotes** are a group of tetrapods whose living members are the reptiles, including birds, and mammals.
- Amniotes are named for the major derived character of the clade, the ***amniotic egg***, which contains membranes that protect the embryo.
- The ***extraembryonic membranes*** *are the* ***amnion, chorion, yolk sac,*** *and* ***allantois***.
- Amniotes have other terrestrial adaptations, such as relatively impermeable skin and the ability to use the rib cage to ventilate the lungs.

The amniotic egg



Reptiles - lay shelled eggs on land

- The **reptile** clade includes the tuataras, lizards, snakes, turtles, crocodilians, birds, and the extinct dinosaurs.
- Reptiles have scales that create a waterproof barrier.
- Most reptiles are **ectothermic**, absorbing external heat as the main source of body heat.
- Birds are **endothermic**, capable of keeping the body warm through metabolism.

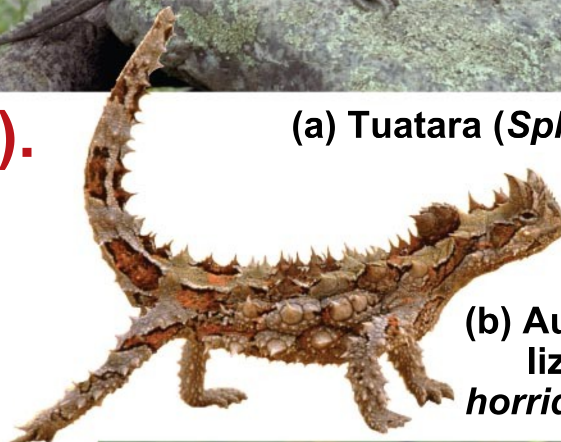
Hatching reptiles

- Dinosaurs diversified into a vast range of shapes and sizes.
- They included bipedal carnivores called **theropods.**
- Paleontologists have discovered signs of parental care among dinosaurs.
- Dinosaurs, with the exception of birds, became extinct by the end of the Cretaceous. Their extinction may have been partly caused by an asteroid.

Extant reptiles

(other than birds).

(a) Tuatara (*Sphenodon punctatus*)



(b) Australian thorny devil lizard (*Moloch horridus*)

(c) Wagler's pit viper (*Tropidolaemus wagleri*)

(d) Eastern box turtle (*Terrapene carolina carolina*)

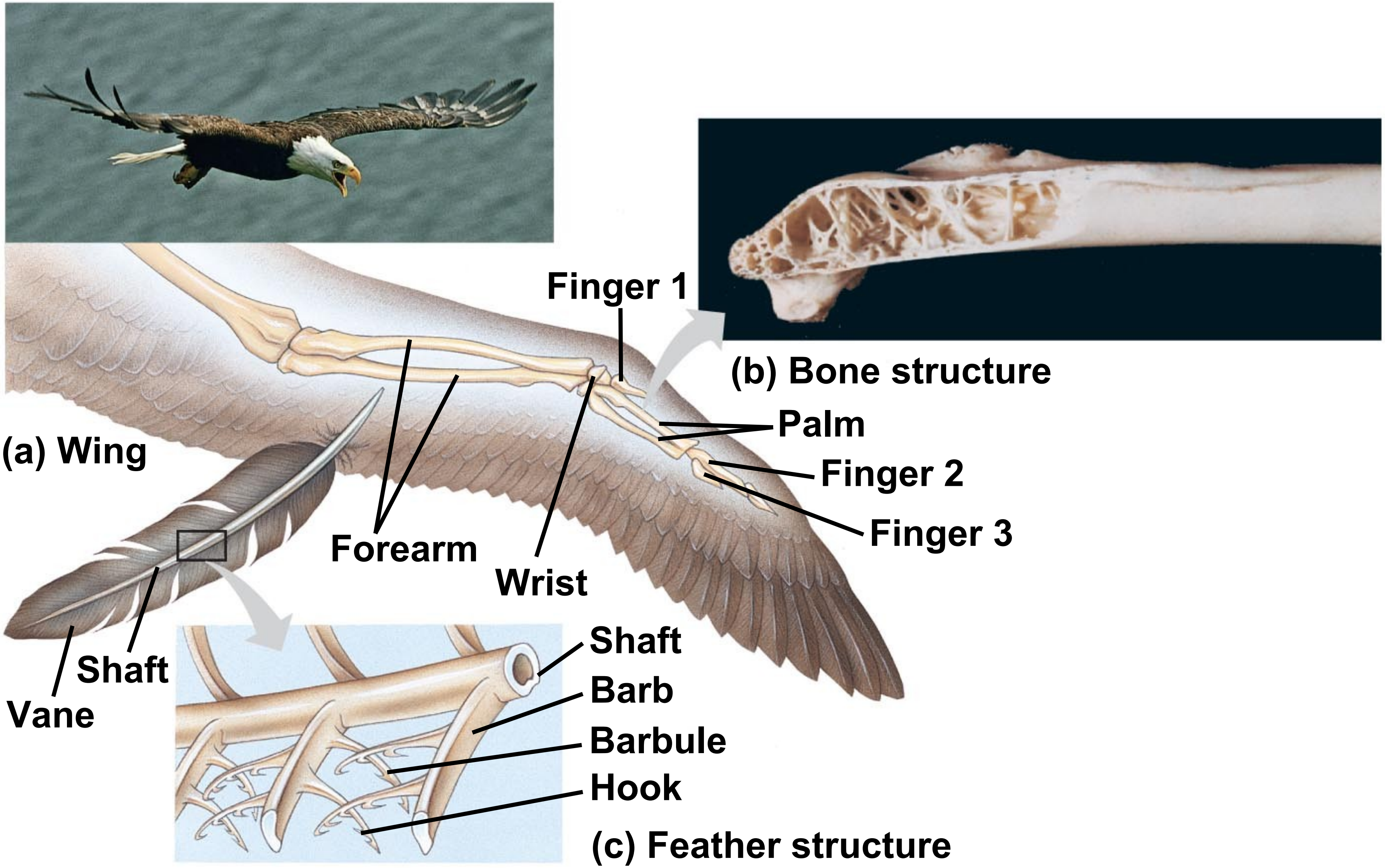
(e) American alligator (*Alligator mississippiensis*)

Birds - reptilian anatomy modified for Flight

Derived Characters of Birds:

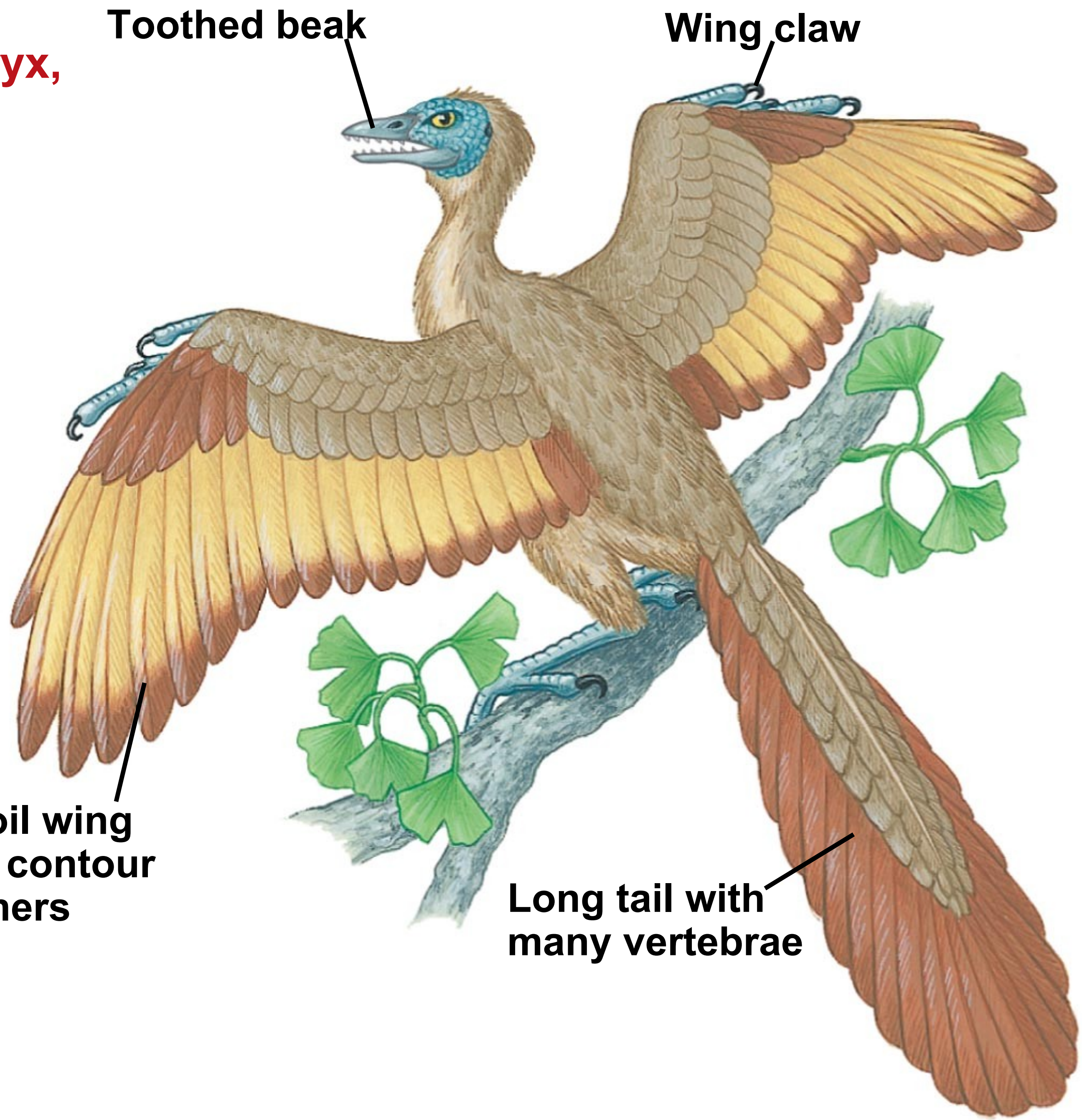
- Many characters of birds are adaptations that facilitate flight
- The major adaptation is wings with keratin feathers
- Other adaptations include lack of a urinary bladder, females with only one ovary, small gonads, and loss of teeth.

Form fits function: the avian wing and feather



- Flight enhances hunting and scavenging, escape from terrestrial predators, and migration.
- Flight requires a great expenditure of energy, acute vision, and fine muscle control.
- Birds probably descended from small theropods, a group of carnivorous dinosaurs. By 150 million years ago, feathered theropods had evolved into birds.
- *Archaeopteryx* remains the oldest bird known.

Archaeopteryx, the earliest known bird



Diversity among living birds

(a) Emu - flightless

(b) Mallards - web feet

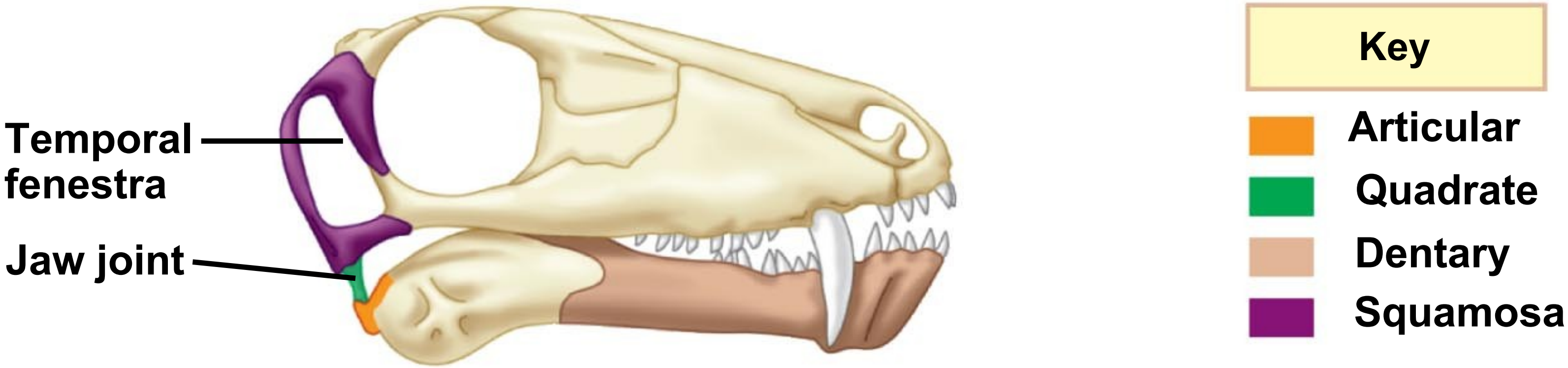
(c) Laysan albatrosses

(d) Barn swallows

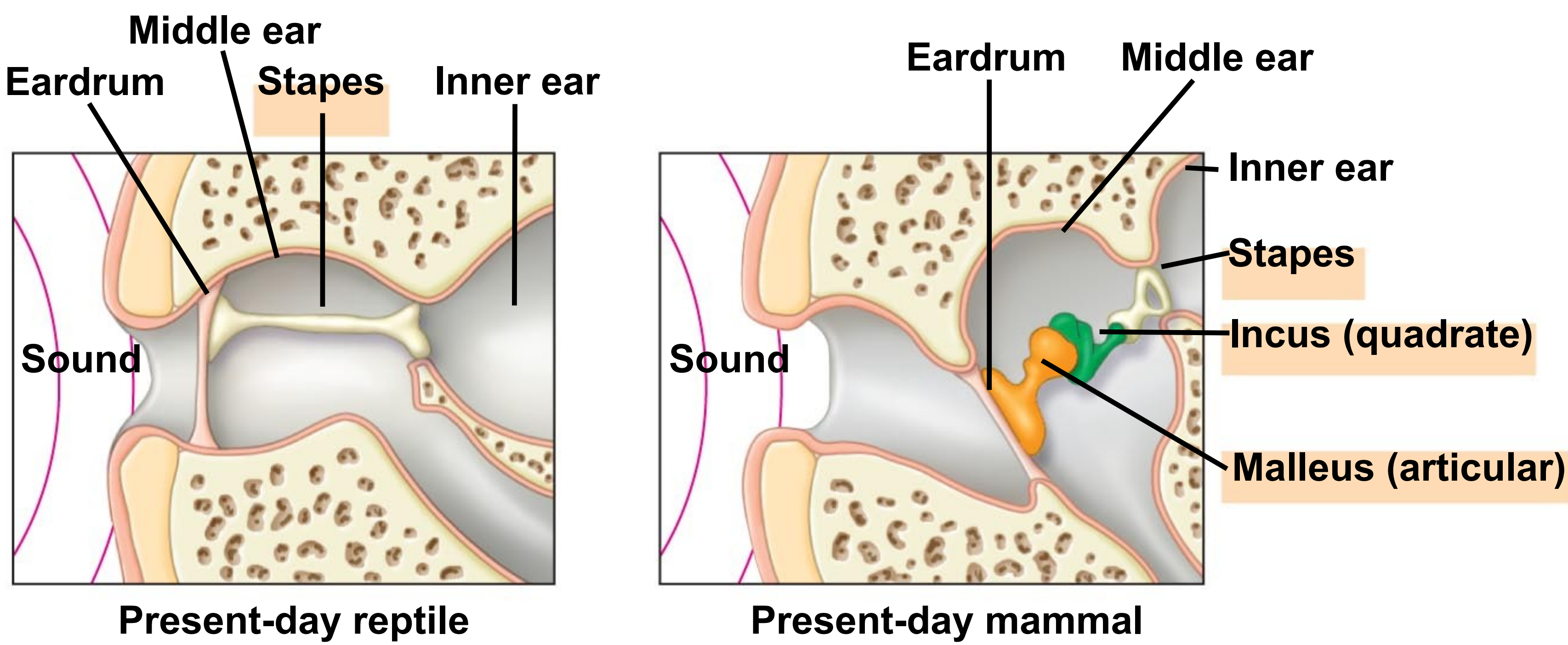
Derived Characters of Mammals

- **Mammals**, class Mammalia, are represented by more than 5,300 species.
- *Mammals have*
  - *Mammary glands, which produce milk*
  - *Hair*
  - *A larger brain than other vertebrates of equivalent size*
  - *Differentiated teeth*.

evolution of the mammalian ear bones



(a) In *Biarmosuchus,* an early synapsid, the articular and quadrate bones formed the jaw joint.



(b) In mammals, the articular and quadrate bones are incorporated into the middle ear.

- By the early Cretaceous, the three living lineages of mammals emerged: monotremes, marsupials, and eutherians.
- **Monotremes** are a small group of egg-laying mammals consisting of echidnas and the platypus.

an Australian monotreme

Marsupials

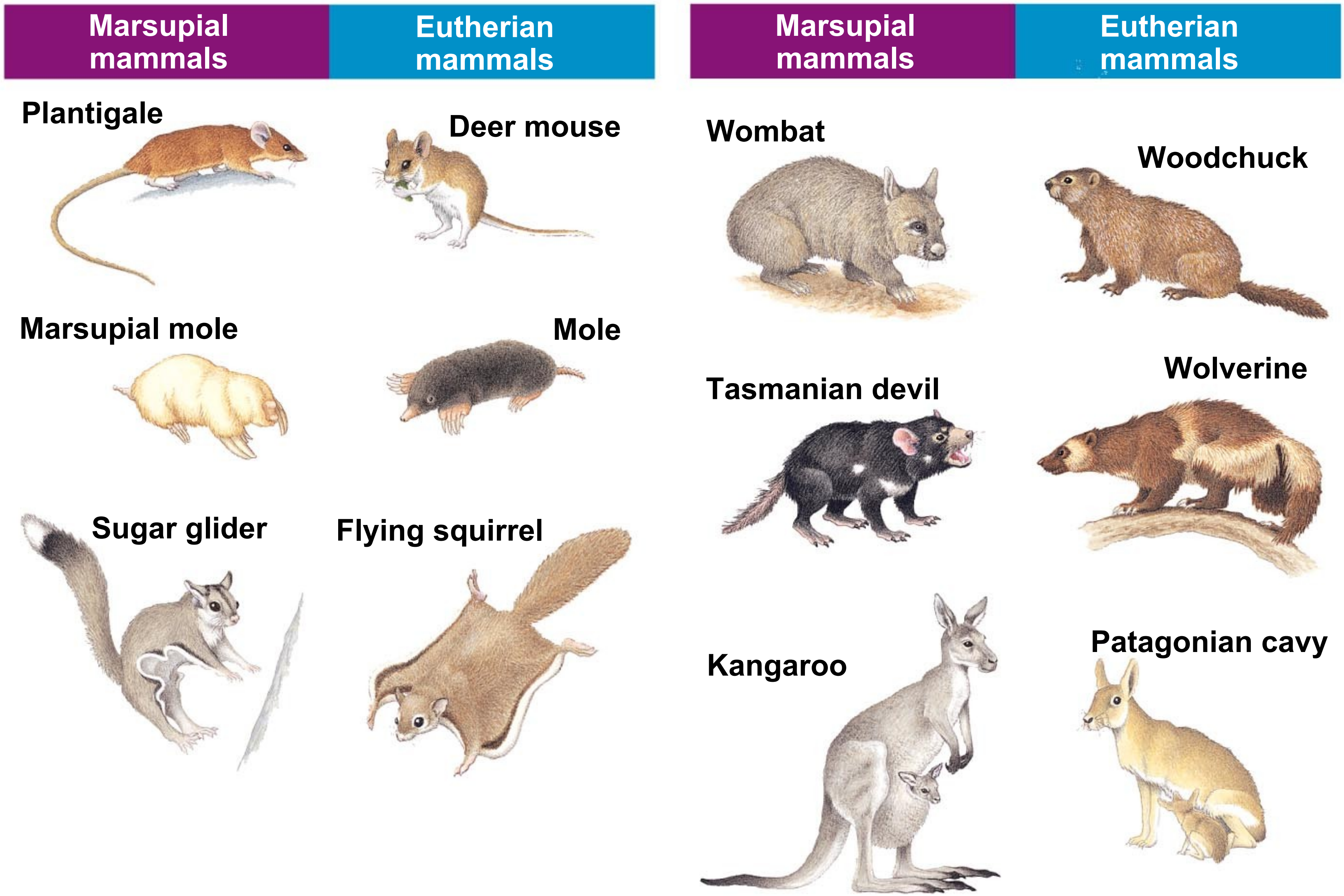
- **Marsupials** include opossums, kangaroos, and koalas.
- The embryo develops within a **placenta** in the mother’s uterus.
- A marsupial is born very early in its development. It completes its embryonic development while nursing in a maternal pouch called a marsupium.

Australian marsupials

(a) A young brushtail possum

(b) Long-nosed bandicoot

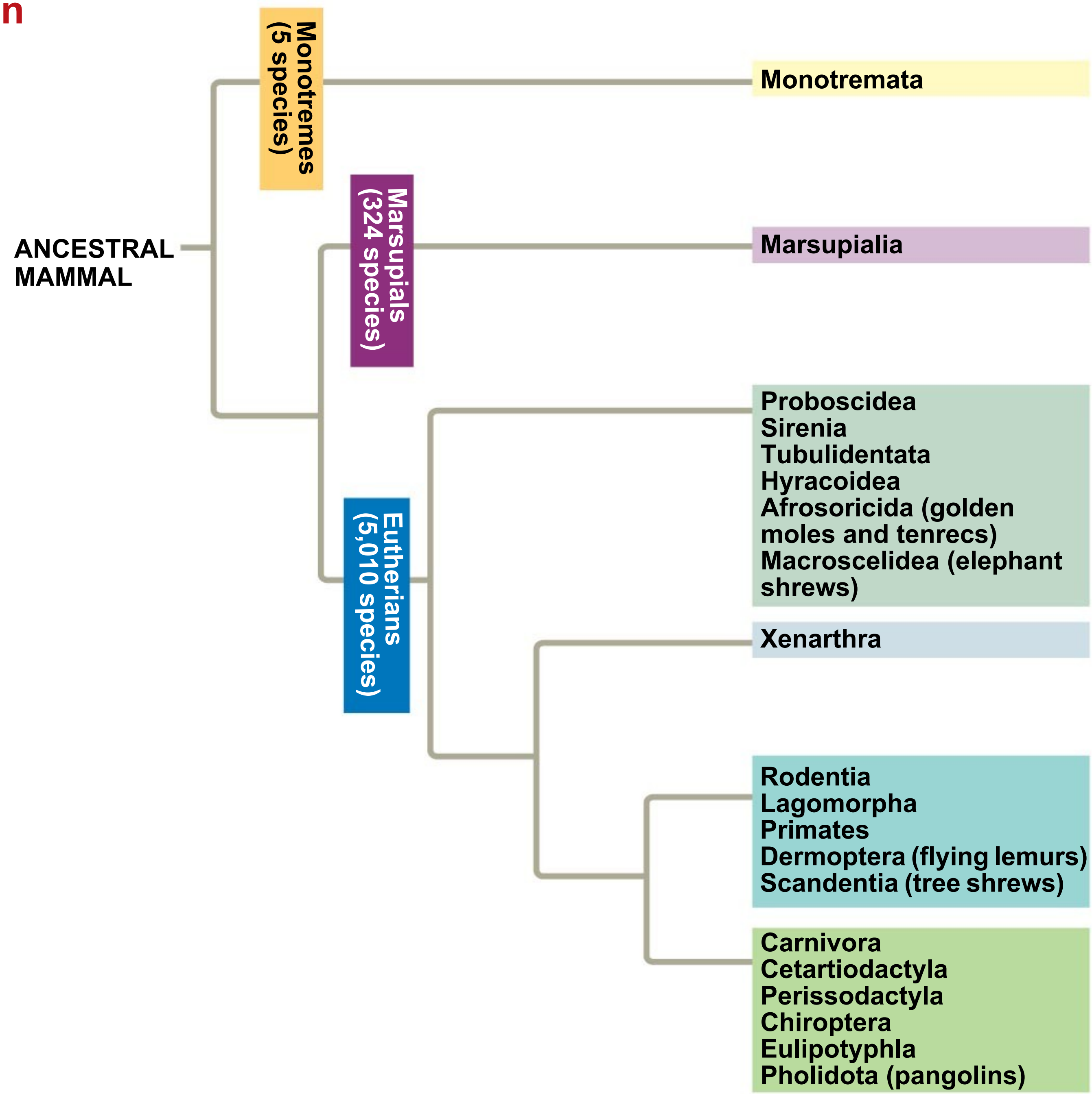
Evolutionary convergence of marsupials and placental mammals



Eutherians - Placental Mammals

- Compared with marsupials, ***eutherians = placental mammals*** have a longer period of pregnancy.
- *Young complete their embryonic development within a uterus, joined to the mother by the placenta*.
- Molecular and morphological data give conflicting dates on the diversification of eutherians.
- In Australia, convergent evolution has resulted in a diversity of marsupials that resemble the eutherians in other parts of the world.

Mammalian Diversity



Mammalian diversity

Orders and Examples	Main Characteristics
Monotremata Platypuses, echidnas  Echidna	Lay eggs; no nipples; young suck milk from fur of mother
Proboscidea Elephants  African elephant	Long, muscular trunk; thick, loose skin; upper incisors elongated as tusks
Sirenia Manatees, dugongs  Manatee	Aquatic; finlike forelimbs and no hind limbs; herbivorous
Xenarthra Sloths, anteaters, armadillos  Tamandua	Reduced teeth or no teeth; herbivorous (sloths) or carnivorous (anteaters, armadillos)
Lagomorpha Rabbits, hares, picas  Jackrabbit	Chisel-like incisors; hind legs longer than forelegs and adapted for running and jumping; herbivorous
Carnivora Dogs, wolves, bears, cats, weasels, otters, seals, walruses  Coyote	Sharp, pointed canine teeth and molars for shearing; carnivorous
Cetartiodactyla **Artiodactyls** Sheep, pigs, cattle, deer, giraffes  Bighorn sheep	Hooves with an even number of toes on each foot; herbivorous
Cetaceans Whales, dolphins, porpoises  Pacific white-sided porpoise	Aquatic; streamlined body; paddle-like forelimbs and no hind limbs; thick layer of insulating blubber; carnivorous

Orders and Examples	Main Characteristics
Marsupialia Kangaroos, opossums, koalas  Koala	Embryo completes development in pouch on mother
Tubulidentata Aardvarks  Aardvark	Teeth consisting of many thin tubes cemented together; eats ants and termites
Hyracoidea Hyraxes  Rock hyrax	Short legs; stumpy tail; herbivorous; complex, multichambered stomach
Rodentia Squirrels, beavers, rats, porcupines, mice  Red squirrel	Chisel-like, continuously growing incisors worn down by gnawing; herbivorous
Primates Lemurs, monkeys, chimpanzees, gorillas, humans  Golden lion tamarin	Opposable thumbs; forward-facing eyes; well-developed cerebral cortex; omnivorous
Perissodactyla Horses, zebras, tapirs, rhinoceroses  Indian rhinoceros	Hooves with an odd number of toes on each foot; herbivorous
Chiroptera Bats  Frog-eating bat	Adapted for flight; broad skinfold that extends from elongated fingers to body and legs; carnivorous or herbivorous
Eulipotyphla "Core insecti-vores": some moles, some shrews  Star-nosed mole	Diet consists mainly of insects and other small invertebrates

Primates

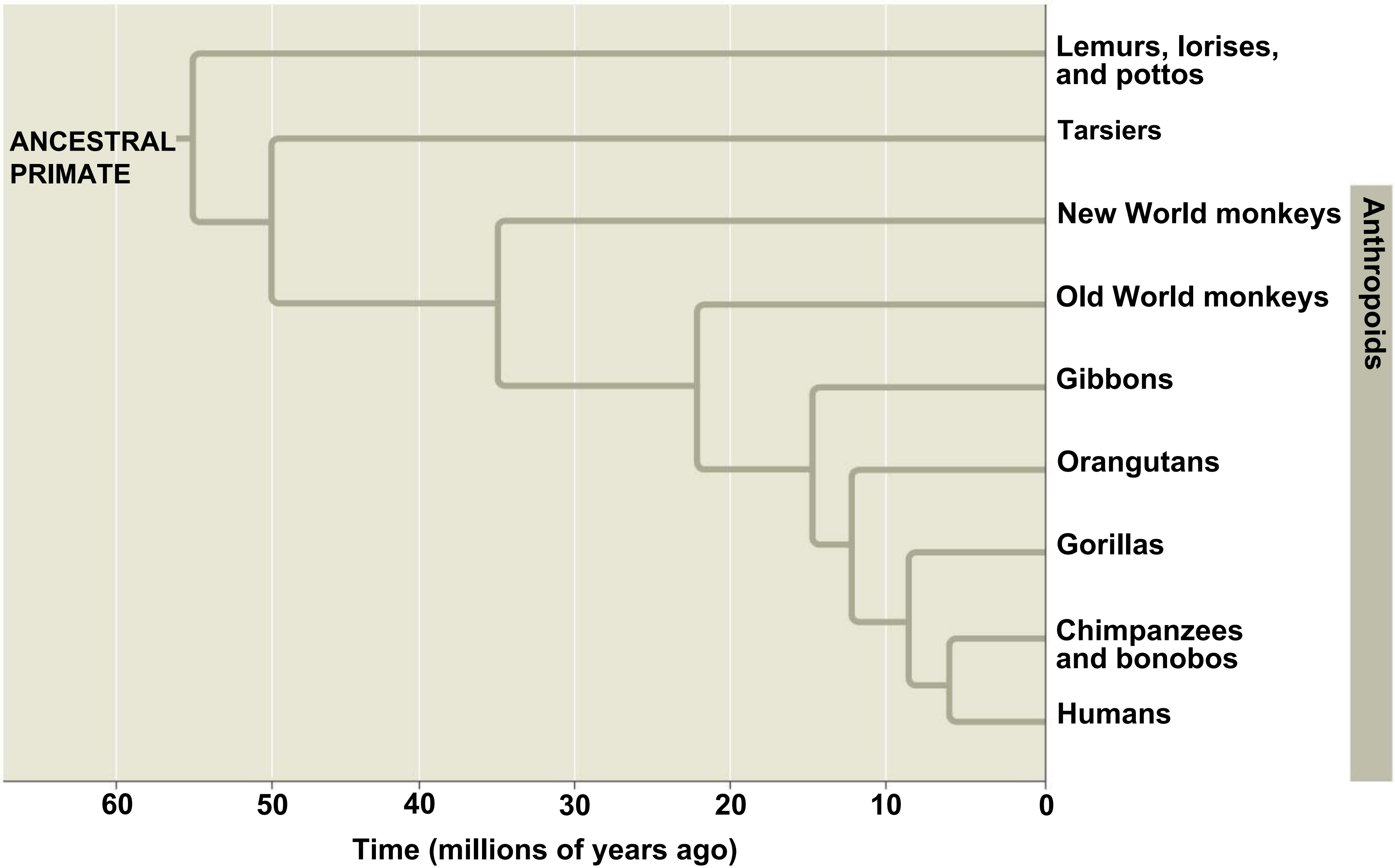
- The mammalian order Primates includes lemurs, tarsiers, monkeys, and apes.
- There are three main groups of living primates:
  - Lemurs, lorises, and pottos
  - Tarsiers
  - **Anthropoids** (monkeys and apes)
- Humans are members of the ape group.

Lemurs

Derived Characters of Primates

- Most primates have hands and feet adapted for grasping.
- Other derived characters of primates:
  - A large brain and short jaws
  - Forward-looking eyes close together on the face, providing depth perception
  - Complex social behavior and parental care
  - A fully **opposable thumb** (in monkeys and apes).

A phylogenetic tree of primates



- The first monkeys evolved in the Old World (Africa and Asia).
- In the New World (South America), monkeys first appeared roughly 25 million years ago.
- New World and Old World monkeys underwent separate adaptive radiations during their many millions of years of separation.

New World monkeys and Old World monkeys

(a) New World monkey

(b) Old World monkey

Nonhuman apes

(a) Gibbon

(b) Orangutan

(c) Gorilla

(d) Chimpanzees

(e) Bonobos

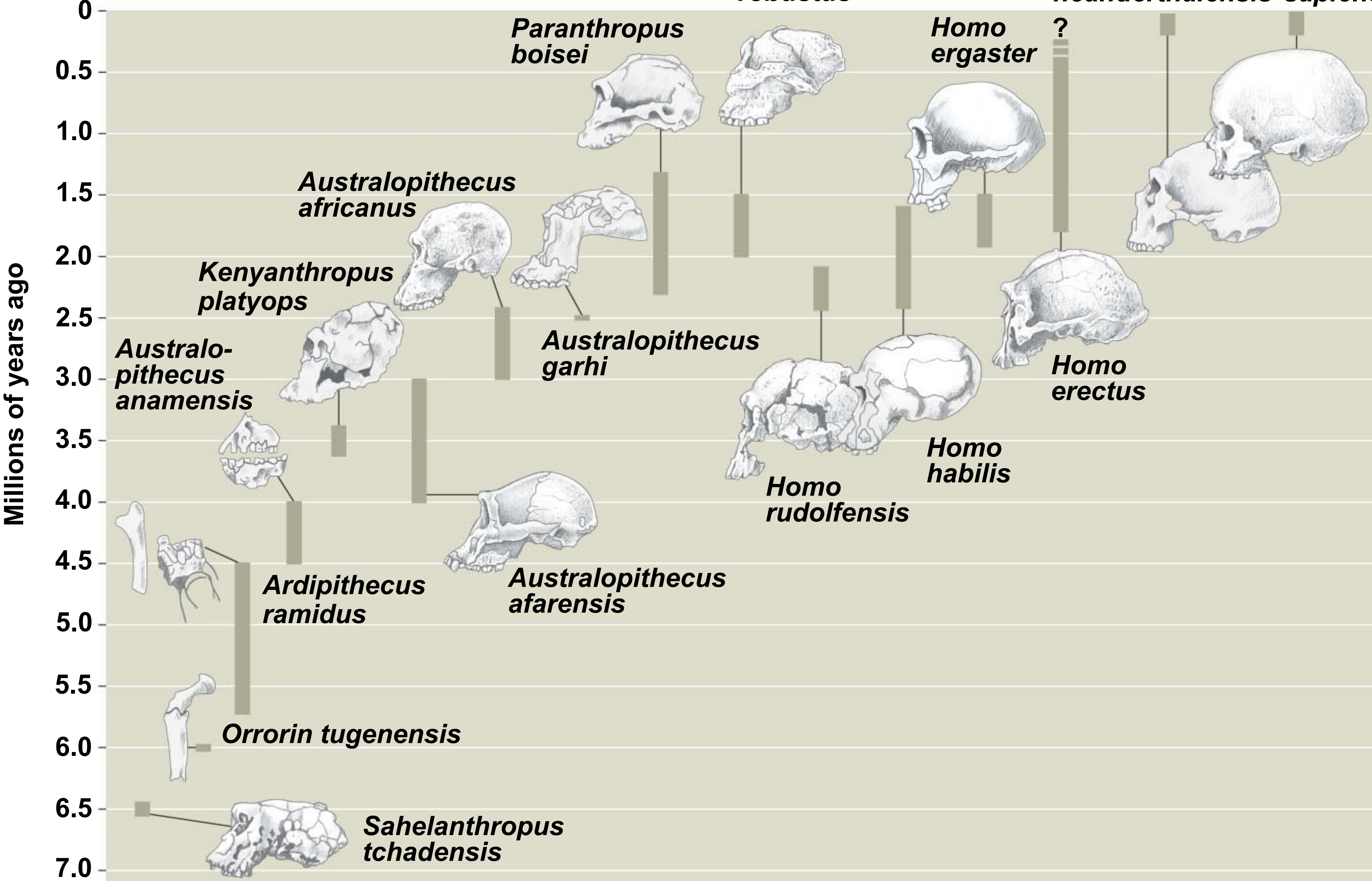
Concept 34.8: Humans are mammals that have a large brain and bipedal locomotion

- The species *Homo sapiens* is about 200,000 years old, which is very young, considering that life has existed on Earth for at least 3.5 billion years.
- The study of human origins is known as **paleoanthropology.**
- **Hominins** (formerly called hominids) are more closely related to humans than to chimpanzees.
- Paleoanthropologists have discovered fossils of about 20 species of extinct hominins.

Derived Characters of Humans

- A number of characters distinguish humans from other apes:
  - Upright posture and bipedal locomotion
  - Larger brains
  - Language capabilities and symbolic thought
  - The manufacture and use of complex tools
  - Shortened jaw
  - Shorter digestive tract.

A timeline for some selected hominin species

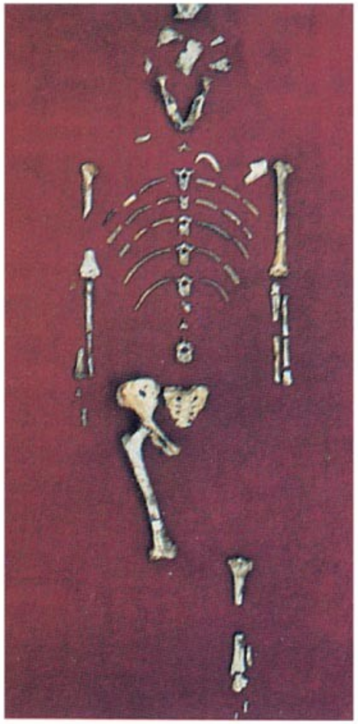


- Hominins originated in Africa about 6–7 million years ago
- Early hominins had a small brain but probably walked upright.
- Two common misconceptions about early hominins:
  - Thinking of them as chimpanzees
  - Imagining human evolution as a ladder leading directly to *Homo sapiens.*

Australopiths

- Australopiths are a paraphyletic assemblage of hominins living between 4 and 2 million years ago.
- Some species walked fully erect.
- “Robust” australopiths had sturdy skulls and powerful jaws.
- “Gracile” australopiths were more slender and had lighter jaws.

Upright posture predates an enlarged brain in human evolution



(a) ***Australopithecus afarensis*** **skeleton**

(b) The Laetoli footprints

(c) An artist's reconstruction of what *A. afarensis* may have looked like

Bipedalism & Tool Use

- Hominins began to walk long distances on two legs about 1.9 million years ago.
- The oldest evidence of tool use, cut marks on animal bones, is 2.5 million years old.
- The earliest fossils placed in our genus *Homo* are those of *Homo habilis*, ranging in age from about 2.4 to 1.6 million years.
- Stone tools have been found with *H. habilis*, giving this species its name, which means “handy man.”

- ***Homo erectus*** originated in Africa by 1.8 million years ago

- It was the first hominin to leave Africa.

- ***Neanderthals***, *Homo neanderthalensis,* lived in Europe and the Near East from 200,000 to 28,000 years ago.

- They were thick-boned with a larger brain, they buried their dead, and they made hunting tools.

Homo Sapiens

- *Homo sapiens* appeared in Africa by 195,000 years ago.
- All living humans are descended from these African ancestors.
- The oldest fossils of *Homo sapiens* outside Africa date back about 115,000 years and are from the Middle East.
- Humans first arrived in the New World sometime before 15,000 years ago.

160,000-year-old fossil of Homo sapiens

- Rapid expansion of our species may have been preceded by changes to the brain that made cognitive innovations possible.
  - For example, the ***FOXP2 gene is essential for human language***, and underwent intense natural selection during the last 200,000 years.
- *Homo sapiens* were the first group to show evidence of symbolic and sophisticated thought.

You should now be able to:

1. List the derived traits for: chordates, craniates, vertebrates, gnathostomes, tetrapods, amniotes, birds, mammals, primates, humans.
2. Describe the trends in mineralized structures in early vertebrates.
3. Describe and distinguish between Chondrichthyes and Osteichthyes.
4. Describe an amniotic egg and explain its significance in the evolution of reptiles and mammals.
5. Explain why the reptile clade includes birds.

6. Distinguish among monotreme, marsupial, and eutherian mammals.

7. Define the term hominin.

8. Describe the evolution of Homo sapiens from australopith ancestors, and clarify the order in which distinctive human traits arose.

9. Explain the significance of the *FOXP2* gene.